Also, all maps stay the same except those now pointing out of i; these are replaced by compositions of the inclusion of $\ker\varphi$ into $\bigoplus_{j\to i} V_j$ with the projections $\bigoplus_{j\to i} V_j \to V_i$.

Definition 6.6.4. Let Q be a quiver, and let $i \in Q$ be a source. Let V be a representation of Q. Let ψ be the canonical map

$$\psi : V_i \to \bigoplus_{i\to j} V_j.$$

Then we define the reflection functor

$$F_i^- : \operatorname{Rep} Q \to \operatorname{Rep} \overline{Q_i}$$

by the rule

$$F_i^-(V)_k = V_k \quad \text{if } k \neq i,$$

$$F_i^-(V)_i = \operatorname{Coker}(\psi) = \left(\bigoplus_{i\to j} V_j\right) / \operatorname{Im}\psi.$$

Again, all maps stay the same except those now pointing into i; these are replaced by the compositions of the inclusions $V_k \to \bigoplus_{i\to j} V_j$ with the natural map $\bigoplus_{i\to j} V_j \to \bigoplus_{i\to j} V_j / \operatorname{Im}\psi$.

Proposition 6.6.5. *Let Q be a quiver and V be an indecomposable representation of Q.*

(1) Let $i \in Q$ be a sink. Then either $\dim V_i = 1$, $\dim V_j = 0$ *for $j \neq i$ or*

$$\varphi : \bigoplus_{j\to i} V_j \to V_i$$

is surjective.

(2) Let $i \in Q$ be a source. Then either $\dim V_i = 1$, $\dim V_j = 0$ *for $j \neq i$ or*

$$\psi : V_i \to \bigoplus_{i\to j} V_j$$

is injective.

Proof. (1) Choose a complement W of $\operatorname{Im}\varphi$. Then we get

$$V \quad = \quad \begin{tikzcd} \bullet & W \arrow[l] \arrow[r] & \bullet \\ 0 & & 0 \\ & \bullet \arrow[u] & \\ & 0 & \end{tikzcd} \quad \oplus \quad V'.$$